



UNIVERSITY OF BARISHAL

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROJECT PROPOSAL

CCTV Based Rescue & Emergency Detection System

CSE-3104 : Software Engineering and information System Design Lab.

Submitted By :

Group No : 05

Abdullah Al Maruf (230102023)

Sadia Mostarin Shuvra (230102032)

Ahasanul Islam Mugdho (230102025)

Deep Paul (230102018)

Antor Sarker (230102021)

Dip Dey (230102038)

Submitted To :

Md Samsuddoha

Assistant Professor, Dept. of CSE, University of Barishal

Date of Submission : 17 May 2026

1. Introduction

Area of Application

The proposed project belongs to the domain of Artificial Intelligence, Computer Vision, Smart Surveillance, and Public Safety Systems. The system will be used in smart cities, hospitals, shopping malls, factories, educational institutions, traffic management systems, and security agencies where continuous surveillance and emergency monitoring are important.

What the System Will Do

The “CCTV Based Rescue & Emergency Detection System” is an AI-powered smart surveillance system that will analyze live CCTV/IP camera feeds in real time using YOLOv8 and OpenCV. The system will automatically detect emergency situations such as:

- Fire and smoke
- Human falls
- Road accidents
- Unauthorized intrusion
- Suspicious crowd behavior

When an emergency is detected, the system will:

- Generate instant alerts
- Save incident snapshots
- Store event details in the database
- Send notifications through email, sound alerts, or phone alerts
- Maintain logs for future review

Intended Users

The intended users of the system are:

- Smart city authorities
- Security agencies
- Hospitals
- Shopping malls
- Factories
- Educational institutions
- Traffic control departments

Importance & Expected Impact

The project is important because traditional CCTV systems mainly record videos and depend heavily on human monitoring. This system can improve emergency response speed, reduce human monitoring limitations, improve public safety, and support intelligent smart surveillance systems. It can also help save lives, reduce property damage, and improve security management.

2. Problem Statement

Limitations of Current Systems / Manual Processes

Traditional CCTV systems depend completely on manual monitoring by security personnel. Human operators must continuously watch multiple camera feeds for long periods, which often causes fatigue, reduced attention, and delayed response during emergencies.

Existing CCTV systems mainly function as passive recording systems. They can record incidents but cannot automatically analyze dangerous situations or generate immediate alerts.

Current systems face several limitations:

- Delayed emergency detection
- Human monitoring errors
- Inefficient multi-camera monitoring
- Lack of intelligent automation
- Slow rescue response
- Increased risk to human life and property

Why the Problem is Significant

The problem is significant because emergencies such as fire outbreaks, road accidents, unauthorized intrusion, human falls, and suspicious crowd behavior require immediate detection and response. Delayed response may cause:

- Loss of human life
- Increased accident damage
- Property destruction
- Public safety risks
- Financial losses

In crowded public places and smart city environments, manually monitoring all surveillance cameras efficiently is nearly impossible. Therefore, an intelligent AI-based emergency detection system is necessary to improve monitoring efficiency and public safety.

3. Motivation

Why We Chose This Idea

We chose this idea because traditional CCTV systems are passive and cannot automatically detect emergency situations. Modern society requires intelligent surveillance systems that can actively monitor environments and provide rapid emergency response.

The increasing number of accidents, fire incidents, and security threats inspired us to develop a smart AI-powered emergency detection system.

Inspiration Behind the Project

The project was inspired by:

- Real-world emergency situations
- Delayed rescue response problems
- Human monitoring limitations
- Growing smart city development
- Advancements in Artificial Intelligence and Computer Vision

Benefits to Users, Organizations, and Society

The system will provide several benefits:

- Faster emergency response
- Improved public safety
- Reduced human monitoring workload
- Better security management
- Reduced accident damage
- Protection of human life and property

Organizations such as hospitals, shopping malls, factories, and traffic departments can improve surveillance efficiency and emergency management using this system.

Potential Improvements & Innovations

This project introduces innovation by combining:

- Artificial Intelligence
- Deep Learning
- Real-time CCTV monitoring
- Automated alert systems
- Database management
- Smart surveillance technology

Future improvements may include:

- Face recognition
- Heatmap analysis
- Weapon detection
- Cloud-based monitoring
- Mobile app integration
- Multi-camera support

4. Expected Features / Requirements

Core Features

- Fire and Smoke Detection
- Human Fall Detection
- Intrusion Detection
- Suspicious Crowd Behavior Detection
- Road Accident Detection
- Real-time Emergency Alerts
- Live CCTV/IP Camera Monitoring
- Incident Snapshot Capture
- Event Logging System
- Email Notification System
- Alert Sound System
- Admin Dashboard
- Rescue Team Panel
- Multi-camera Support
- Database Management System

5. Development Approach

Selected Development Model

The project will follow the Agile SDLC Model.

Why Agile Instead of Waterfall

Although the Waterfall model is a traditional software development approach, this project requires continuous AI model improvement, testing, debugging, and feature updates. Therefore, Agile is more suitable because it provides flexibility and iterative development.

Development Process

1. Requirement Analysis
2. System Planning
3. System Design

4. Frontend and Backend Development
5. AI Model Integration
6. Database Integration
7. Testing and Debugging
8. Deployment
9. Maintenance and Future Improvement

6. Tools & Technology

Application Type

- Web-Based Application

Frontend Technologies

- HTML
- CSS
- JavaScript
- Tailwind CSS

Backend Technologies

- Python
- Flask Framework

AI & Computer Vision Technologies

- OpenCV
- YOLOv8
- PyTorch

Database

- MySQL

Video Streaming Technologies

- OpenCV Streaming
- WebRTC

Development Tools

- VS Code
- GitHub
- Google Colab

7. Budget Estimation

Item	Cost
CCTV / IP Cameras & Sensors	₹4,00,000
Server & Cloud Infrastructure	₹3,00,000
AI Model Development & Training	₹2,50,000
Software Development	₹2,00,000
Networking & Internet Setup	₹1,00,000
Testing & Maintenance	₹1,00,000
Miscellaneous Expenses	₹1,50,000
Total Estimated Cost	₹15,00,000

8. Deployment & Maintenance

Deployment

The system will be deployed on a normal PC or server without using Raspberry Pi or embedded hardware. CCTV cameras, IP cameras, or webcams will be connected for real-time monitoring.

Maintenance

Regular maintenance will include:

- Bug fixing
- AI model optimization
- Security updates
- Database maintenance
- Performance improvement

Future upgrades may include:

- Face recognition
- Cloud monitoring
- Advanced analytics
- Mobile application support

9. Timeline

Task	Duration
Requirement Analysis & Planning	Week 1
System Design & Architecture	Week 1
AI Model Development	Week 2
Frontend & Backend Development	Week 3
Database Integration	Week 4
Testing & Debugging	Week 4
Deployment & Documentation	Week 5

10. SDLC

Selected SDLC Model

The project will follow the Agile SDLC Model.

SDLC Phases

1. Requirement Analysis

Identify system requirements such as fire detection, intrusion detection, emergency alerts, and live CCTV monitoring.

2. System Design

Design the system architecture, database schema, dashboard interface, and AI detection workflow.

3. Development

Develop frontend pages, Flask backend, database integration, and AI detection modules using YOLOv8 and OpenCV.

4. Testing

Test AI detection accuracy, alert generation, video streaming, and database functionality.

5. Deployment

Deploy the system on a PC/server and connect CCTV or IP cameras.

6. Maintenance & Improvement

Fix bugs, improve AI performance, and add future features.

11. Conclusion

The CCTV Based Rescue & Emergency Detection System is an AI-powered smart surveillance solution designed to improve public safety and emergency response using Artificial Intelligence and Computer Vision technologies.

The system can automatically detect dangerous situations such as fire, accidents, intrusion, and suspicious activities in real time and instantly notify responsible authorities. This reduces human monitoring limitations, improves emergency response efficiency, and enhances security management.

The project has strong practical and social impact because it can help save lives, reduce accident damage, protect property, and support smart city development. Future enhancements such as face recognition, cloud-based monitoring, and advanced analytics can make the system more powerful and scalable.